Development and Evaluation of a Personalized Kanji Learning System Using Multiple Example Sentences

Kotaro Matsushita Hideo Suzuki
Tokyo University of Information Sciences
4-1 Onaridai, Wakaba, Chiba 265-8501, Japan
{matusita, suzuki}@rsch.tuis.ac.jp

Takahiro Kurumagawa
Sanwa Solutions Co., Ltd
36-10 Sakae, Chuo, Chiba 264-0016, Japan
g13001tk.edu@gmail.com

Takuya Sakiyama
JMC Co., Ltd.
1-8-8 Nakameguro, Meguro, Tokyo 153-0061, Japan
g10082ts@live.jp

Noriko Uosaki
Osaka University
1-1 Yamadaoka, Suita 565-0871, Japan
uosaki@ciee.osaka-u.ac.jp

***Abstract*— This paper describes the development and evaluation of a user adaptive *kanji* learning system for JFL (Japanese as a foreign language) learners. Learning *kanji* is a real challenge for learners of Japanese as a foreign language. Since the context is important in the process of language learning, multiple example sentences to provide various contexts where the target *kanji* is used were implemented. In order to enhance its usability, English navigation was introduced. The result of the pilot evaluation showed that the average score increased in terms of its usability, while it revealed some user unfriendliness in *kanji* display. Our future works include its improvement such as addition of *kanji* meaning links and keyboard control.**

Keywords*— *e-Learning; JFL (Japanese as a Foreign Language); Kanji; Personalized Learning System

I. INTRODUCTION

The progress of computing technology and network environment has enabled us to provide learning opportunities such as multimedia learning, *technology-enhanced learning* (TEL), *computer-based instruction* (CBI), which allow students to use PCs, tablets, or mobile devices on one-to-one or one-to-group basis. The development of e-Learning systems and researches on their practical uses have been very active accordingly.

We have developed user adaptive learning systems since 2012[1][2][3]. In our developed system, once the users select the contents that they want to learn, it automatically generates quizzes from the selected contents.

However, it has long been pointed out that the context is very important in language learning. Taking the contexts into consideration, we have developed *kanji* learning system with multiple example sentences to provide various contexts where the target *kanji* is used. Binary-choice quizzes are automatically generated using multiple example sentences so that they can learn how the target *kanji* is used in different contexts. By using multiple example sentences, it is expected to avoid them to memorize the sentence used in the quiz. The pilot experiment was conducted in order to evaluate the effectiveness of our system.

The rest of the paper includes the system outline featuring its refinements as well as the result of the pilot evaluation.

II. PERSONALIZED KANJI LEARNING SYSTEM

A. System outline

As mentioned, the system automatically generates random quizzes from the contents that the learners selected. They can select *kanji* across school years. They can select either *kunyomi* (Japanese original reading) or *onyomi* (similar to Chinese reading, but different). The computer graphics animations were created using the POV-Ray software to show whether the answer they choose is correct or not. In this study, English navigation was introduced and multiple example sentences were implemented to provide different quiz sentences in each target *kanji*.

B. System construction and screen structure

The system is a client-server application, which runs on the web browser. It is programmed using HTML, JavaScript and PHP. The computer graphics were created using the POV-Ray software. Figs. 1 and 2 show the system constructure and the screen structure.

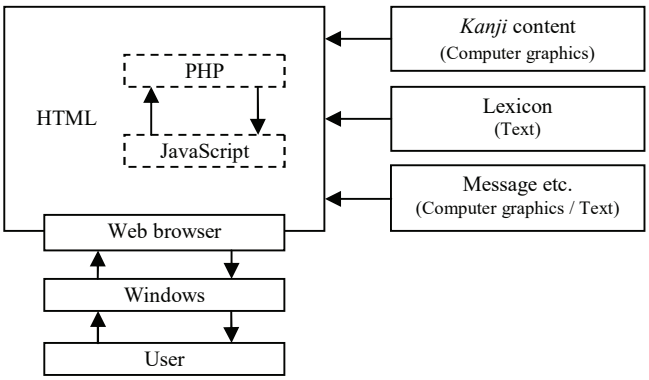

Fig.1 System constructure

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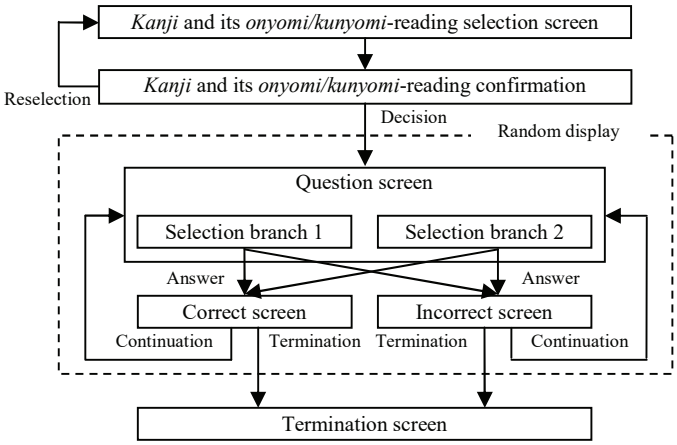
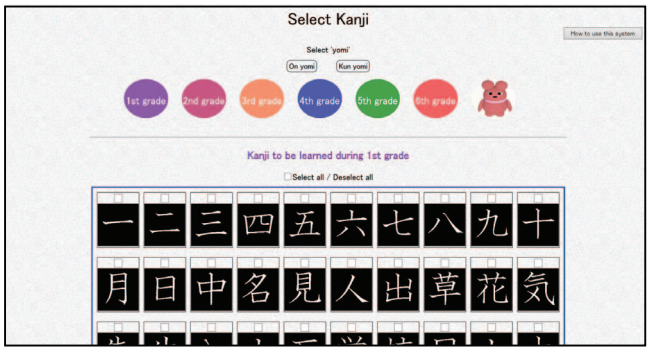


Fig.2 Screen structure

C. System operation

First the system displays the *kanji* and its *onyomi/kunyomi*-reading selection screen (Fig. 3(a)). On this screen, learners select *kanji* characters and *onyomi* or *kunyomi* readings that they intend to learn. On the *kanji* and its *onyomi/kunyomi*-reading confirmation screen (Fig. 3(b)), they checks if the selected *kanji* and its *onyomi/kunyomi* readings are completely correct. When they click the start button, two-choice quizzes are generated by the system (Fig. 3(c)).

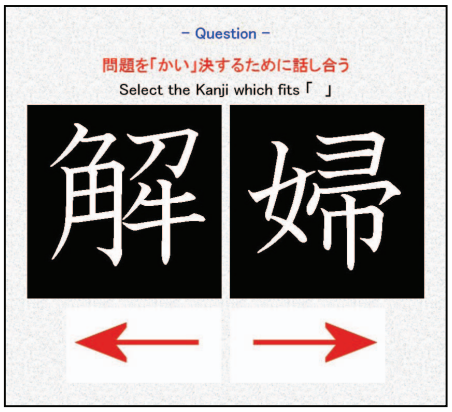


(a) *Kanji* and its *onyomi/kunyomi*-reading selection screen

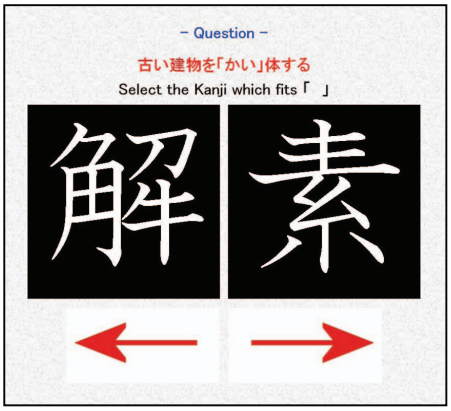
(b) *Kanji* and its *onyomi/kunyomi*-reading confirmation screen

Multiple example sentences are randomly choosen to use as quiz sentences. They read a question and select *kanji* which they think is correct. After they answer the question, the correct or incorrect answer screen shows the corresponding computer graphics animations (Fig. 3(d), (e), (f), (g)).

Sentence example 1



Sentence example 2



Sentence example 3

(c) Question screen（multiple example sentence）

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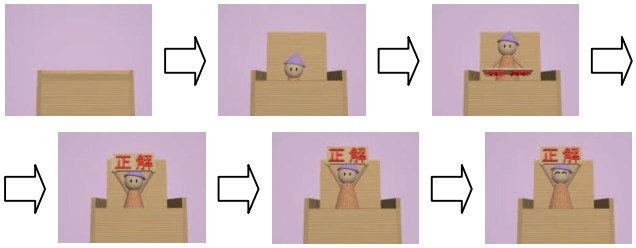
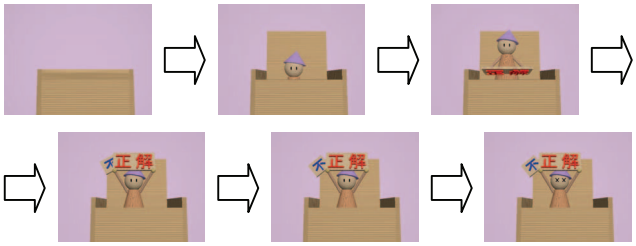
(d) Correct answer screen

(e) Incorrect answer screen



(f) Computer graphics animations on correct answer screen



(g) Computer graphics animations on incorrect answer screen

Fig.3 Screen transition

III. EVALUATION OF THE SYSTEM

A. Procedures

Sixteen international students of CALL(computer assisted language learning) class held weekly at Osaka University participated in the evaluation. After receiving the briefing about how to use the system, they were assigned to use it both during class and out-of-class as a home assignment for one week. After the evaluation, a questionnaire survey was conducted to obtain their feedback. Table 1 shows the questions asked in the questionnaire survey.

Table.1 Questions used in the five-point-scale questionnaire

- Was it easy for you to use *kanji* Learning System? (Question1)
- Was it helpful for your *kanji* learning? (Question 2)
- Was it fun for you to use the system? (Question 3)
- Did you like the CG animation? (Question4)

B. Results and Discussion

Table 2 shows the mean and SD (standard deviation) values of five-point-scale questionnaire. The highest score of 3.75 was given when asked whether it was easy to use the system. Since English navigation was introduced, the average score was higher than that of our previous study (3.69) [2]. The rest of average scores was lower than those of our previous study (Mean:3.10 vs. 3.28). Table 3 shows their free comments. Some user-unfriendliness of the system as shown in Table 3 was pointed out.

Table.2 The result of the questionnaire

$n = 16$	Question1	Question2	Question3	Question4
Average	3.75	2.44	3.13	3.12
SD	1.25	1.37	1.17	1.31

Table.3 Free comments (excerpts)

- I think i would improve if you can view the reading of each *kanji* in order to study.
- Meaning of the *kanjis* would be nice to know.
- hope there is any explanation of the sentences will be better.
- Keyboard controls would be nice, using keys for the choices is more convenient than having to move the mouse over it.
- *Kanji* that do not have a Kun-Yomi should be skipped automatically.
- The animations should be an option that can be turned off. Animations are nice as a motivational tool for new learners, but loses its novelty very quickly and only slows down the process once people review 100 or more *kanji*.

IV. CONCLUSION

In this paper we have described the use of the refined *kanji* Learning System with English navigation and multiple example sentences for facilitating international students to learn *kanji*. The result of the pilot evaluation revealed that the system improved its usability, but some user-unfriendliness was found. Our future works include its improvement such as an addition of *kanji* meaning links and keyboard control.

REFERENCES

[1] K. Matsushita, T. Sakiyama, Y. Fujisawa, T. Kurumagawa, H. Koyama, H. Suzuki. K.J. Mackin, E. Nunohiro, "Development of a User Adaptive Graphical Learning System", Proceedings of the 21st International Conference on Computers in Education, pp.463-465, November 2013.

[2] K.Matsushita, T.Kurumagawa, N.Nemoto, T.Sakiyama, H.Suzuki, K.J.Mackin, Y.Nagai, E.Nunohiro, "Development and Evaluation of a User Adaptive Kanji Learning System Using Computer Graphics", Proceedings of the 22nd International Conference on Computers in Education, pp.385-387, November 2014.

[3] N.Uosaki, K.Matsushita, H.Suzuki,"Supporting JFL learners with User Customized Kanji Learning System Using Computer Graphics", Proceedings of the 4th International Congress on Advanced Applied Informatics, pp.264-267 , July 2015.

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